Waterbench: Towards Democratizing Secure Image Authorship with Watermarking

Fanni Büki
Ludwig Maximilian University
Munich, Germany
f.bueki@campus.lmu.de

Eric Echtermeyer
Ludwig Maximilian University
Munich, Germany
eric.echtermeyer@campus.lmu.de

David Hoffmann
Ludwig Maximilian University
Munich, Germany
david.hoffmann@campus.lmu.de

***Abstract*—As AI-generated images become increasingly realistic, safeguarding content authenticity and ownership has emerged as a critical challenge for individual users. While numerous watermarking techniques exist, their practical suitability for end users remains largely unexplored. This work presents the first systematic benchmarking of post-processing watermarking methods specifically focused on accessibility, evaluating robustness, image quality, and computational feasibility across five open-source algorithms. Our evaluation across eight attack types and 100 images per configuration reveals that MBRS offers the best practical trade-off, achieving high compression robustness (0.94 Ratcliff score) while maintaining processing times suitable for consumer hardware (0.42 seconds per image). We identify a critical weakness across all evaluated methods: poor geometric robustness, with scores below 0.28, indicating watermarks cannot survive common transformations like rotation and cropping. To democratize watermark technology, we developed Waterbench, making invisible watermarking accessible to non-technical users.**

***Index Terms*—Image Watermarking, Watermark Robustness, Benchmark Evaluation, Digital Authentication, Post-Processing Methods**

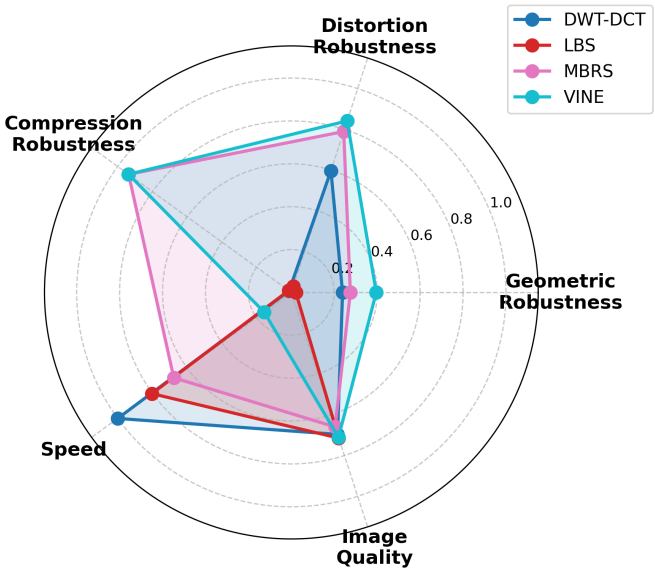

Fig. 1. Comparison of spatial- (LSB [6]), transformation- (DWT-DCT [7] and deep-learning based (RAW, [10], MBRS [9], VINE [11]) watermarking methods across five key dimensions: (1) Lossy JPEG compression, (2) distortion robustness (e.g. brightness, contrast, blurring and noising), (3) geometric attacks (e.g. rotation, scale and crop), (4) image quality after watermarking (imperceptibility) and (5) speed of the watermarking method on consumer hardware. All metrics are averaged across message lengths and attack magnitudes.

I. Introduction

The growing ability to generate and digitally manipulate images at a low cost in conjunction with the wide spread use of digital image sharing platforms raise the need for robust mechanisms to prove authenticity and ownership. End users share images on social media platforms where they are distributed making the ability to prove authorship of an image an important part of digital sovereignty. This can be addressed with digital watermarking techniques which embed an invisible message into the bits of an image file. While a plethora of image watermarking methods exist [1]–[11], to the best of our knowledge, these are never evaluated with the specific aim of usability and efficacy for private users. In this work we benchmark a wide pallet of invisible watermarking methods with the requirements of end users in mind.

> *How can we democratize cryptographically-verified image ownership through accessible watermarking technology?*

To address this, the fundamental trade-off between watermark robustness and computational demands has to be reconciled while maintaining image quality. A visible watermark overlaying the image clearly degrades it's quality. In contrast, the invisible watermarking techniques we benchmark, make subtle changes to the bits of the digital image. However, this can still lead to noise and artifacts which we want to keep minimal to maintain imperceptibility of the watermark. Some methods [8], [11] include expensive computations such as diffusion to embed watermarks which are then more robust to compression or editing, while other methods [6], [7] rely on simple embedding techniques that are less robust.

We systematically evaluate open-source watermarking algorithms in terms of robustness, image quality and computational demands to identify techniques suitable for accessible deployment. While VINE achieves the highest overall robustness, its processing time of 1.81 seconds per image makes it impractical for real-time applications. MBRS emerges as

the most practical solution for end users, balancing strong compression robustness (0.94 Ratcliff score) with acceptable processing speeds (0.42 seconds per image) on consumer hardware. Critically, all evaluated methods struggle with geometric attacks, indicating that watermark survival under rotation and cropping remains a fundamental challenge.

To further push for the democratization of image ownership we develop Waterbench, an interactive platform that makes cryptographically-verified image ownership accessible to anyone with a web browser. Beyond ownership claims, Waterbench serves as both a research instrument and an educational resource through systematic benchmarking of watermarking algorithms and interactive evaluation modes.

Our contributions are summarized as follows:

- We conduct the first end user oriented benchmark of invisible watermarking methods aimed at assessing the trade-off between watermark robustness, image quality, and computational demands.
- We find that MBRS offers a good compromise with high robustness while being excecutable on consumer devices.
- Our Waterbench (waterbench.org) application makes invisible watermarking accessible to non-technical users.

II. Watermarking Methods

Digital image watermarking is a content protection technique that embeds information covertly into host images. An effective watermarking scheme must balance three often competing requirements: *robustness*, *computational demands* and *image quality*.

Techniques can be broadly classified into **in-processing** techniques [1]–[5] which add watermarking during the artificial generation process of an image and **post-processing** techniques [6]–[11] which can be applied to any digital image. In line with our goal of focusing on end users we only consider methods of the latter category. These can be further classified by their embedding domain into spacial transform-based and deep learning methods.

A. Classical Watermarking Techniques

Spatial Domain methods, such as LSB manipulation [6], directly modify pixel values. These techniques are straightforward to implement and offer high capacity but are extremely *fragile*, failing under basic attacks like compression or filtering. Despite their limitations, these methods serve as an important baseline for understanding watermark robustness.

Transform Domain methods increase robustness by embedding watermarks in frequency coefficients, distributing information across the signal. The Discrete Cosine Transform (DCT) [12] exploits energy compaction properties to embed watermarks in perceptually less sensitive mid-frequency coefficients, offering resilience against filtering and JPEG compression. The Discrete Wavelet Transform (DWT) [13] decomposes images into sub-bands, providing excellent spatial-frequency localization. Hybrid approaches combining DWT, DCT, and Singular Value Decomposition (SVD) represent the classical state-of-the-art, leveraging the stability of singular values against perturbations to achieve superior robustness [7]. However, these classical methods often require external feature matching techniques (e.g., SIFT [14]) to resist complex geometric attacks like rotation and scaling.

B. Deep Learning Approaches

Deep learning approaches represent the current state-of-the-art in watermarking research. These methods typically employ an autoencoder architecture with an encoder, a noiser layer simulating attacks, and a decoder, trained end-to-end to jointly optimize imperceptibility and robustness. The key innovation lies in adversarial training through noiser layers that simulate various attacks during training, forcing networks to learn resilient, non-linear embedding patterns.

StegaStamp [8] pioneered learned steganographic algorithms for robust encoding and decoding of bitstrings into photos. The system demonstrates robust 56-bit error-corrected payload embedding that survives physical printing and re-photography through training on differentiable image augmentations that approximate real-world distortions. StegaStamp's encoder-decoder framework with spatial perturbation modeling established foundational principles for deep watermarking.

MBRS [9] specifically addresses JPEG compression robustness, a critical weakness in prior deep watermarking frameworks. Traditional differentiable JPEG simulation fails to capture the complexity of real JPEG compression's quantization process. MBRS introduces a novel training strategy that randomly switches between real JPEG compression, simulated JPEG, and noise-free layers across mini-batches during training.

RAW [10] introduces a paradigm shift from traditional encoder-decoder architectures. Rather than embedding fixed binary codes in latent representations, RAW embeds learnable watermarks directly into original image data in both frequency and spatial domains. The framework employs a jointly-trained classifier for watermark detection, offering model-agnostic compatibility and plug-and-play deployment. RAW achieves zero-bit capacity detection with provable guarantees on false positive rates even under adversarial attacks. However, the approach of learning watermark does not allow for custom messages or copyright information, making it infeasible for our use-case.

VINE [11] addresses watermark vulnerability to advanced image editing enabled by large-scale text-to-image diffusion models. Through the W-Bench benchmark, VINE demonstrated that most existing watermarking methods fail after image regeneration and various editing operations. VINE leverages pretrained diffusion model priors and identifies that blurring distortions exhibit similar frequency characteristics to image editing, enabling their use as surrogate attacks during training.

III. Benchmarking Framework

The research community has developed several benchmarking frameworks to systematically evaluate watermarking robustness and image quality.

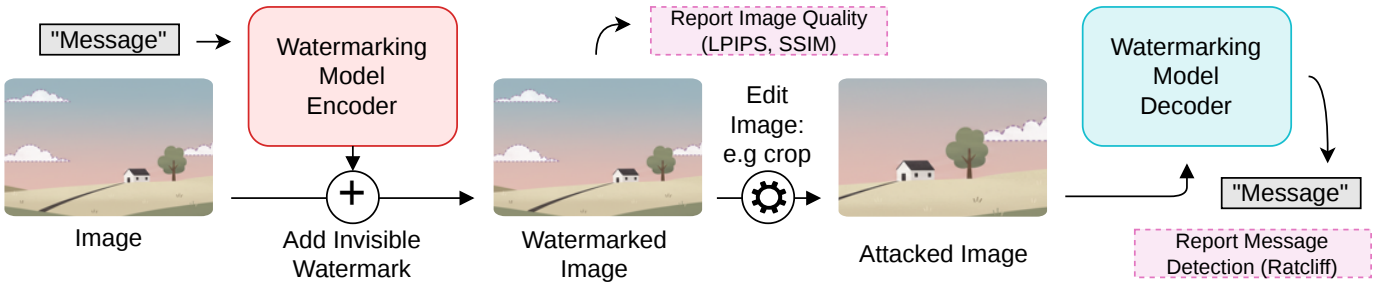

Fig. 2. Our benchmarking pipeline follows a three step process given an image: (1) encode a message as an invisible watermark in the image using one of the methods discussed in section II, (2) compute image quality metrics on the watermarked image to assess its imperceptibility, (3) attack the watermarked image with a transformation (e.g. cropping the image) (4) use the methods decoder to extract the message from the attacked watermarked image and (5) compute watermark detection metrics.

Watermark Analysis Via Enhanced Stress-testing (WAVES) [15] is a standardized evaluation protocols with a taxonomy of attacks that encompasses classical distortions (blurring, rotation, cropping, etc.) and powerful, novel variations of regeneration and adversarial attacks, against watermarks. While the framework evaluates the same dimensions we are interested in, they mainly focus on computationally demanding deep learning models.

W-Bench [11] is introduced with VINE as the first benchmark which evaluates watermark robustness to image regeneration, global editing, local editing and image-to-video generation. The authors evaluate a broad set of models also including non-deep learning based methods and our work can be seen as an extension and validation of their results with democratization of watermarking in mind.

Specifically, our work evaluate two methods that where not compared in [11]: the spacial LSB method and the deep learning based RAW model. Furthermore, the discussion and placement of our results is geared towards usability for private end users and not mainly focused on advancing the state-of-the-art.

In our benchmarking we evaluate five models covering all embedding domains and broadly representing the spectrum from established classical methods to recent state of the art approaches. We include LSB [6] as a spatial model and important baseline, DWT-DCT [7] as a transform based techniques and MBRS [9], RAW [10] and VINE [11] as deep learning based approaches.

A. Image Quality and Imperceptibility

To evaluate the imperceptibility of watermarks we measure similarity between the original and embedded images containing the watermark with two different metrics: Learned Perceptual Image Patch Similarity (LPIPS) [16] and Structural Similarity Index Measure (SSIM) [17].

SSIM is using degradation of structural information to capture image quality. The metric ranges from -1 to 1, with value of 1 indicating perfect similarity between the reference and distorted images. It is defined as

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (1)$$

where μ_x and μ_y denote the mean intensities of image patches x and y, σ_x^2 and σ_y^2 their variances, and σ_{xy} their covariance. The constants C_1 and C_2 are included to stabilize the division for small denominators.

LPIPS computes the perceptual distance between two images by comparing deep feature activations extracted from a pre-trained network. This measure aligns well with human perception by capturing high-level features and textures in images rather than focusing strictly on structural elements. The distance is computed as a weighted sum of squared differences between normalized feature maps across multiple layers. The value of the metric ranges from 0 to 1. A low LPIPS score indicates higher perceptual similarity. It is defined as

$$d(x, x_0) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} ||w_l \odot (\hat{y}^l_{hw} - \hat{y}^l_{0,hw})||_2^2 \quad (2)$$

where l indexes the selected layers of a pre-trained convolutional neural network, y^l and y^l_0 denote the feature activations of images x and x_0 at layer l, and $\hat{y}^l$ and $\hat{y}^l_0$ are the channel-wise ℓ_2-normalized feature maps. H_l and W_l denote the spatial height and width of the feature maps at layer l, and w_l represents learned channel-wise weighting coefficients.

B. Robustness Attacks

The embedded images are then subjected to eight different attacks which we group into three categories: (1) distortion attacks which change the visual appearance of the image content, (2) compression attacks which apply lossy compression and (3) geometric attacks which rotate, scale or crop the image. A detailed breakdown of all attacks is presented in Table II in the appendix.

C. Message reconstruction quality

To compare the original and extracted messages, we utilize the Ratcliff/Obershelp Similarity score [18] which compares

the two strings by doubling the sum of all matching subsequences and dividing by the total length of both strings. It is defined as

$$R_{\mathrm{O}} = \frac{2|K_m|}{|S_1| + |S_2|} \tag{3}$$

where S_1 and S_2 denote the two strings being compared, $|S_n|$ represents the length of string n, and $|K_m|$ denotes the sum of the lengths of all common substrings found by the recursive algorithm. The constant 2 is a scaling factor used to account for the length being counted relative to the total length of both strings. The resulting score ranges from 0 to 1, where 1 indicates identical strings.

An important difference two other matching scores such as the Jaccard index which treat the two sequences as a bag of words, the Ratcliff/Obershelp similarity score is sensitive to the position and sequence of characters, making it a more sophisticated measure.

Note that as a zero-bit embedding approach the RAW method, cannot be evaluated using matching but instead gives a match accuracy for the learned watermark.

IV. Experimental Results

In the following we apply the benchmarking framework as described in section III for LSB [6], DWT-DCT [7], MBRS [9], RAW [10] and VINE [11]. Each of the methods is evaluated on different message sizes from 2 to 30 bytes (2, 4, 8, 15, 30 ASCII characters). Note that the resulting experiments reach the limits of MBRS and VINE which only support 30 and 12 characters respectively. These two are evaluated up to the supported number of characters. For further implementation details refer to our code at: https://github.com/echtermeyer/Data-Security.

The benchmarking results reveal substantial differences in how watermarking methods reconstruct embedded messages under various attacks. Figure 3 shows that the deep learning methods VINE and MBRS exhibit high overall robustness across most attack types. The classical method DWT-DCT achieves moderate performance, while LSB exhibits low performance even under minimal image modifications. This behaviour comes from LSB's dependence on least significant bits, which the evaluated attack types drastically alter.

Table I shows that all methods are fairly adequate in terms of imperceptibility (image quality) as measured through LPIPS and SSIM, with MBRS exhibiting the lowest quality score. LSB achieves perfect quality which is expected, since the change of the least significant bits is not captured by the extracted local statistical parameters. All remaining methods yield close to optimal results for both metrics. A detailed breakdown by message length is provided in Table III in the appendix, revealing that image quality remains consistent across message lengths for all evaluated methods.

VINE and MBRS perform particularly well against filter-like attacks that preserve image geometry. Brightness adjustments, noise, contrast changes, scaling, and JPEG compression

Method	LPIPS	SSIM
DWT-DCT	0.10 ± 0.07	0.96 ± 0.01
LBS	0.00 ± 0.00	1.00 ± 0.00
MBRS	0.23 ± 0.06	0.88 ± 0.05
RAW	0.03 ± 0.02	0.94 ± 0.02
VINE	0.01 ± 0.01	0.99 ± 0.00

TABLE I
LPIPS and SSIM (mean ± pooled std) averaged across all message lengths.

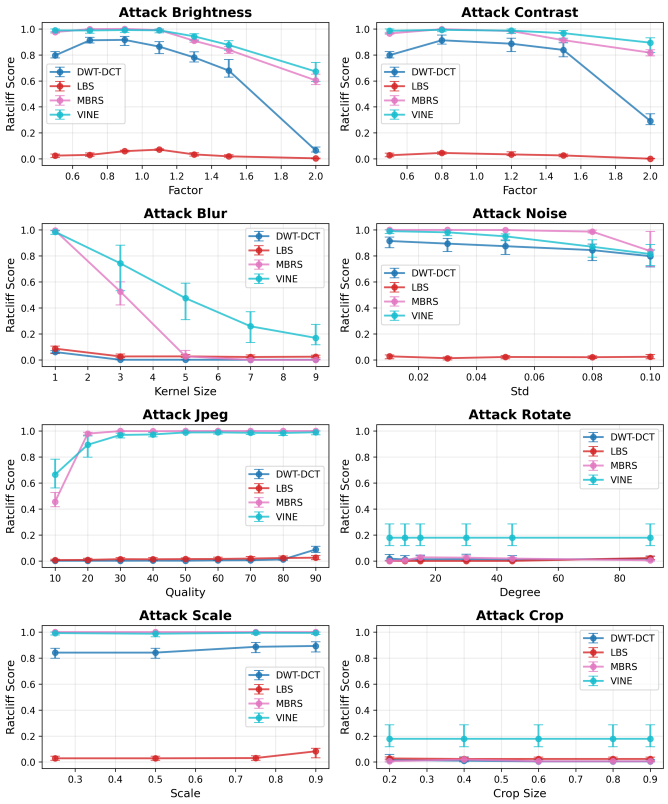

Fig. 3. Watermark robustness across different attack types and magnitudes. Data points represent the mean Ratcliff score averaged across all message lengths and images. Error bars indicate the range (min–max) across different message lengths. Eight attack categories are evaluated: brightness and contrast adjustments, Gaussian blur with varying kernel sizes, additive Gaussian noise, JPEG compression at different quality levels, rotation angles, scaling factors, and center cropping.

all yield high reconstruction scores, with MBRS achieving 0.79 for distortion attacks and 0.94 for compression attacks. However, geometric attacks such as rotation and cropping cause significant failures, with geometric robustness scores of only 0.28 for MBRS and 0.40 for VINE. This limitation carries practical implications: watermarks likely cannot survive screenshotting, which typically involves cropping portions of the original image. The poor performance on geometric attacks probably reflects their underrepresentation in the training data, presenting an ongoing challenge even for these powerful

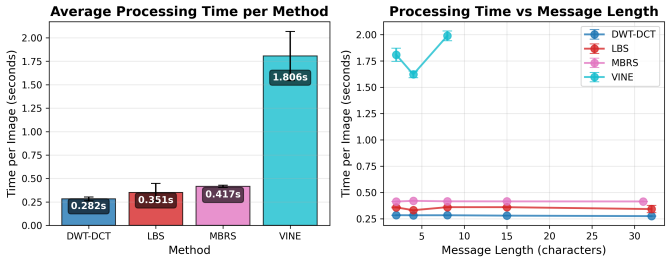

Fig. 4. Processing time comparison across watermarking methods. **Left:** Average time per image (in seconds) for each method, computed as the mean across all attack configurations and message lengths. Error bars show standard deviation. **Right:** Processing time versus message length, showing how encoding time scales with the number of characters in the watermark. Each data point represents the mean time for a specific message length. Both plots aggregate data from the complete benchmark suite of 8 attack types with varying magnitudes. Note that results for VINE are only reported up to the maximum supported message length of 12 characters.

methods.

Beyond reconstruction quality, processing time directly impacts user experience and determines practical feasibility. Figure 4 presents average processing times across methods and message lengths.

VINE requires 1.81 seconds per image on average, which is substantially longer than competing methods. This represents a critical limitation when compared to MBRS, which achieves similar robustness at 0.42 seconds per image. Notably, processing time remains roughly constant across message lengths, allowing methods to embed and extract messages of any supported length in constant time.

To have a direct comparison, we define five key dimensions for watermarking performance:

- **Distortion Robustness:** Mean Ratcliff score across brightness, contrast, blur, and noise attacks.
- **Geometric Robustness:** Mean Ratcliff score across rotation, scaling, and cropping attacks.
- **Compression Robustness:** Mean Ratcliff score across JPEG compression at various quality levels.
- **Image Quality:** Mean SSIM and mean LPIPS between watermarked and original images across all configurations.
- **Speed:** Relative processing efficiency, normalized such that the fastest method scores 1.0 and slower methods receive proportionally lower scores.

Figure 1 illustrates each method's strengths and weaknesses across these dimensions. VINE demonstrates the best overall performance with distortion robustness of 0.84 and compression robustness of 0.94, while MBRS achieves 0.79 and 0.94 respectively. However, VINE's prohibitively high processing times make MBRS the most promising candidate for real-time watermarking applications among the tested methods.

The RAW method was excluded from the primary comparison as it implements a zero-bit detection rather than embedding full messages. Thus, it can only determine if a watermark is present without carrying any other ownership information that is required for cryptographic claims. Nevertheless, RAW achieves moderate detection accuracy (0.81 mean across all attacks) while maintaining acceptable image quality as shown in Table III. The performance remains consistent across message lengths, as expected for detection-only methods. Similar to message-based approaches, RAW shows strong robustness in terms of accuracy against distortion attacks (brightness: 0.97, noise: 0.98) but struggles with geometric transformations (rotation: 0.20). However, as binary detection is fundamentally simpler than extracting multi-bit messages, these accuracy scores are not directly comparable to the Ratcliff reconstruction scores of message-based methods.

V. Waterbench

As a complement to our benchmarking analysis, we introduce **Waterbench** (waterbench.org), an application designed to increase accessibility for claiming and verifying authorship while providing an educational tool for understanding watermarking robustness. The platform splits these functionalities into the *Ownership Mode* and the *Challenge Mode*.

The *Ownership Mode* establishes mathematically verifiable proof of authorship through public-key cryptography:

1) **Identity Creation:** Users generate an RSA-2048 keypair on their first login. The browser securely stores this keypair, while the backend registry records the public key for verification purposes.
2) **Claim Generation:** The user's private key cryptographically signs a claim package containing the author's identity, timestamp, and optional message. This process occurs entirely client-side. The system then embeds the signature and claim data invisibly into the image using LSB, creating unfalsifiable proof of possession at a specific date and time.
3) **Verification:** Any party can upload a claimed image to the platform. The API extracts the embedded claim and signature, retrieves the registered public key, and cryptographically verifies the signature against the claim data. Successful verification proves that the image remains unaltered since signing and confirms ownership by the holder of the associated private key.

The *Challenge Mode* enables interactive, real-time robustness testing. Users upload an image, embed a watermark, and then apply common transformations such as distortion, compression, or geometric modifications that simulate real-world attacks. The application provides immediate feedback on watermark survival rates, offering hands-on experience with algorithm vulnerabilities and strengths.

This combination of secure authorship claims and interactive robustness testing positions the platform as both a research instrument and an educational resource.

VI. Limitations & Future Work

Computational complexity represents the primary limiting factor of this work. Among post-processing methods, we were unable to benchmark computationally expensive deep learning approaches like StegaStamp [8]. Additionally, in-processing

watermarking methods such as Stable Signature [19] and Tree-Ring [20], which embed watermarks during the diffusion generation process itself, fall outside the scope of our post-processing benchmark framework. This constraint of computational complexity also extends beyond model selection and impacts the entire evaluation pipeline. Our benchmark only evaluates 100 images with simple attacks, yet the complete benchmarking process spans multiple days, making more comprehensive evaluation infeasible. Additionally, we conduct all benchmarking on consumer hardware running concurrent background processes, which may introduce variability in the reported processing times.

Beyond computational constraints, our attack simulation presents limited realism. The benchmark applies only simple attacks (cropping, rotation, brightness adjustment) in isolation, without examining complex interactions between multiple simultaneous attacks. We exclude screenshotting from our evaluation due to setup limitations, though cropping serves as an approximate proxy since screenshots typically capture only portions of the full image. However, Figure 3 reveals that all evaluated methods already perform poorly under minimal cropping, suggesting that screenshotting represents an even more destructive attack. Future work should investigate more realistic attack scenarios, including compound attacks and screenshotting, to better characterize watermark robustness in real-world environments.

VII. Conclusion

This work systematically evaluates watermarking methods for accessible image ownership verification. Our benchmarking across eight attack types reveals that MBRS offers the best trade-off between robustness and computational efficiency for consumer hardware, while VINE achieves higher robustness at higher processing times. Classical methods like DWT-DCT show mediocre performance, and LSB fails even under minimal perturbations. Critically, all evaluated methods struggle with geometric attacks, particularly cropping and rotation, indicating that robust geometric invariance remains an unsolved challenge. To bridge the gap between research and practical deployment, we developed Waterbench, providing cryptographically verified ownership claims and interactive robustness testing for end users.

VIII. AI Usage Statement

We hereby declare that the report submitted is our own work. We used Gemini 3 Pro [21] and Claude Sonnet 4.5 [22] for spell checking and rephrasing throughout the entire document. Both models where further used to create code snippets for visualization and other analyses.

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APPENDIX

The appendix provides supplemental data for the benchmarking evaluation. Table II documents the specific attack configurations used for the robustness attacks. Furthermore, Table III shows the image quality metrics in detail across varying message lengths. The Accuracy and Ratcliff/Obershelp scores (for RAW and the remaining methods respectively) are presented in Table IV, Table V, Table VI, Table IX, Table VI, Table VII, Table VIII, Table X and Table XII.

Robustness	Attack	Description	Setting
Distortion	Brightness	Adjusts luminosity using the `PIL.ImageEnhance.Brightness`.	Factors: 0.7, 0.9, 1.1, 1.3, 1.5, 2.0
	Contrast	Scales pixel separation using `PIL.ImageEnhance.Contrast`.	Factors: 0.5, 0.8, 1.2, 1.5, 2.0
	Blurring	Convolves image with a Gaussian blur filter of radius k.	Kernel size (k): 1, 3, 5, 7, 9
	Noise	Adds Gaussian noise $\mathcal{N}(0, \sigma)$ to normalized tensor; clamped to $[0, 1]$.	Std Dev (σ): 0.01, 0.03, 0.05, 0.08, 0.1
Compression	JPEG	Simulates artifacts by saving/reloading image buffers.	Quality: 10, 20, 30, 40, 50, 60, 70, 80, 90
Geometric	Rotation	Rotates image counter-clockwise by θ degrees.	Degrees: 5, 10, 15, 30, 45, 90
	Scale	Downsamples then upsamples via Lanczos to original size.	Scale factor: 0.25, 0.5, 0.75, 0.9
	Crop	Crops from top-left, then resizes crop to original size (Lanczos).	Crop factor: 0.2, 0.4, 0.6, 0.8, 0.9

TABLE II
DETAILED BREAKDOWN OF WATERMARK ROBUSTNESS ATTACKS. GEOMETRIC ATTACKS (SCALE, CROP) UTILIZE LANCZOS RESAMPLING TO MAINTAIN THE ORIGINAL IMAGE RESOLUTION ($H \times W$) FOR CONSISTENT BENCHMARKING. THE IMPLEMENTATION IS BASED ON W-BENCH [11].

Method	**Message length**									
	2		4		8		15		30	
	Image quality score									
	LPIPS	SSIM	LPIPS	SSIM	LPIPS	SSIM	LPIPS	SSIM	LPIPS	SSIM
DWT-DCT	0.09 ± 0.06	0.97 ± 0.01	0.09 ± 0.06	0.96 ± 0.01	0.11 ± 0.07	0.96 ± 0.01	0.11 ± 0.07	0.96 ± 0.01	0.11 ± 0.07	0.96 ± 0.01
LBS	0.00 ± 0.00	1.00 ± 0.00	0.00 ± 0.00	1.00 ± 0.00	0.00 ± 0.00	1.00 ± 0.00	0.00 ± 0.00	1.00 ± 0.00	0.00 ± 0.00	1.00 ± 0.00
MBRS	0.24 ± 0.06	0.87 ± 0.05	0.24 ± 0.06	0.87 ± 0.05	0.23 ± 0.06	0.88 ± 0.05	0.23 ± 0.06	0.88 ± 0.05	0.22 ± 0.07	0.88 ± 0.05
RAW	0.03 ± 0.02	0.94 ± 0.02	0.03 ± 0.02	0.94 ± 0.02	0.03 ± 0.02	0.94 ± 0.02	0.03 ± 0.02	0.94 ± 0.02	0.03 ± 0.02	0.94 ± 0.02
VINE	0.01 ± 0.01	0.99 ± 0.00	0.01 ± 0.01	0.99 ± 0.00	0.01 ± 0.01	0.99 ± 0.00	-	-	-	-

TABLE III
LPIPS AND SSIM (MEAN ± STD) ACROSS METHODS AND MESSAGE LENGTHS. **NO ATTACK.**

Method	Message length									
	2		4		8		15		30	
	Similarity score									
	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy
DWT-DCT	0.75 ± 0.29	-	0.71 ± 0.27	-	0.71 ± 0.27	-	0.72 ± 0.27	-	0.69 ± 0.26	-
LBS	0.03 ± 0.09	-	0.03 ± 0.08	-	0.03 ± 0.07	-	0.04 ± 0.07	-	0.04 ± 0.06	-
MBRS	0.91 ± 0.18	-	0.91 ± 0.18	-	0.90 ± 0.16	-	0.90 ± 0.15	-	0.89 ± 0.14	-
RAW	-	0.97 ± 0.10	-	0.97 ± 0.10	-	0.97 ± 0.10	-	0.97 ± 0.10	-	0.97 ± 0.10
VINE	0.89 ± 0.20	-	0.93 ± 0.12	-	0.94 ± 0.10	-	-	-	-	-

TABLE IV
RATCLIFF/OBERSHELP AND ACCURACY (MEAN ± STD) ACROSS METHODS AND MESSAGE LENGTHS. **BRIGHTNESS ATTACK**.

Method	Message length									
	2		4		8		15		30	
	Similarity score									
	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy
DWT-DCT	0.01 ± 0.04	-	0.01 ± 0.04	-	0.01 ± 0.04	-	0.01 ± 0.04	-	0.02 ± 0.04	-
LBS	0.02 ± 0.06	-	0.03 ± 0.06	-	0.03 ± 0.05	-	0.04 ± 0.05	-	0.05 ± 0.05	-
MBRS	0.33 ± 0.13	-	0.32 ± 0.10	-	0.32 ± 0.09	-	0.29 ± 0.07	-	0.29 ± 0.06	-
RAW	-	0.92 ± 0.07	-	0.92 ± 0.07	-	0.92 ± 0.07	-	0.92 ± 0.07	-	0.92 ± 0.07
VINE	0.41 ± 0.15	-	0.55 ± 0.16	-	0.62 ± 0.12	-	-	-	-	-

TABLE V
RATCLIFF/OBERSHELP AND ACCURACY (MEAN ± STD) ACROSS METHODS AND MESSAGE LENGTHS. **BLUR ATTACK**.

Method	Message length									
	2		4		8		15		30	
	Similarity score									
	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy
DWT-DCT	0.79 ± 0.30	-	0.74 ± 0.29	-	0.73 ± 0.29	-	0.75 ± 0.28	-	0.71 ± 0.27	-
LBS	0.02 ± 0.05	-	0.02 ± 0.05	-	0.02 ± 0.05	-	0.03 ± 0.05	-	0.04 ± 0.04	-
MBRS	0.94 ± 0.17	-	0.94 ± 0.16	-	0.93 ± 0.15	-	0.93 ± 0.14	-	0.93 ± 0.13	-
RAW	-	0.49 ± 0.18	-	0.49 ± 0.18	-	0.49 ± 0.18	-	0.49 ± 0.18	-	0.49 ± 0.18
VINE	0.93 ± 0.17	-	0.98 ± 0.06	-	0.98 ± 0.05	-	-	-	-	-

TABLE VI
RATCLIFF/OBERSHELP AND ACCURACY (MEAN ± STD) ACROSS METHODS AND MESSAGE LENGTHS. **CONTRAST ATTACK**.

Method	Message length									
	2		4		8		15		30	
	Similarity score									
	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy
DWT-DCT	0.00 ± 0.00	-	0.00 ± 0.00	-	0.00 ± 0.01	-	0.01 ± 0.02	-	0.03 ± 0.03	-
LBS	0.01 ± 0.02	-	0.02 ± 0.03	-	0.02 ± 0.03	-	0.03 ± 0.03	-	0.04 ± 0.02	-
MBRS	0.01 ± 0.04	-	0.01 ± 0.02	-	0.01 ± 0.03	-	0.01 ± 0.03	-	0.01 ± 0.03	-
RAW	-	0.97 ± 0.05	-	0.97 ± 0.05	-	0.97 ± 0.05	-	0.97 ± 0.05	-	0.97 ± 0.05
VINE	0.13 ± 0.00	-	0.12 ± 0.00	-	0.29 ± 0.00	-	-	-	-	-

TABLE VII
RATCLIFF/OBERSHELP AND ACCURACY (MEAN ± STD) ACROSS METHODS AND MESSAGE LENGTHS. **CROP ATTACK**.

Method	Message length									
	2		4		8		15		30	
	Similarity score									
	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy
DWT-DCT	0.01 ± 0.06	-	0.01 ± 0.04	-	0.01 ± 0.04	-	0.01 ± 0.02	-	0.02 ± 0.03	-
LBS	0.01 ± 0.02	-	0.01 ± 0.02	-	0.02 ± 0.02	-	0.02 ± 0.02	-	0.03 ± 0.02	-
MBRS	0.93 ± 0.06	-	0.93 ± 0.06	-	0.94 ± 0.05	-	0.94 ± 0.04	-	0.95 ± 0.04	-
RAW	-	0.93 ± 0.19	-	0.93 ± 0.19	-	0.93 ± 0.19	-	0.93 ± 0.19	-	0.93 ± 0.19
VINE	0.90 ± 0.17	-	0.94 ± 0.11	-	0.97 ± 0.07	-	-	-	-	-

TABLE VIII
RATCLIFF/OBERSHELP AND ACCURACY (MEAN ± STD) ACROSS METHODS AND MESSAGE LENGTHS. **JPEG QUALITY ATTACK**.

Method	Message length									
	2		4		8		15		30	
	Similarity score									
	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy
DWT-DCT	0.91 ± 0.26	-	0.87 ± 0.26	-	0.87 ± 0.26	-	0.88 ± 0.26	-	0.80 ± 0.31	-
LBS	0.01 ± 0.01	-	0.02 ± 0.02	-	0.02 ± 0.02	-	0.03 ± 0.02	-	0.03 ± 0.01	-
MBRS	0.94 ± 0.07	-	0.95 ± 0.06	-	1.00 ± 0.01	-	0.95 ± 0.05	-	0.98 ± 0.02	-
RAW	-	0.98 ± 0.08	-	0.98 ± 0.08	-	0.98 ± 0.08	-	0.98 ± 0.08	-	0.98 ± 0.08
VINE	0.87 ± 0.21	-	0.94 ± 0.11	-	0.96 ± 0.10	-	-	-	-	-

TABLE IX
RATCLIFF/OBERSHELP AND ACCURACY (MEAN ± STD) ACROSS METHODS AND MESSAGE LENGTHS. **NOISE ATTACK**.

Method	Message length									
	2		4		8		15		30	
	Similarity score									
	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy
DWT-DCT	0.00 ± 0.01	-	0.00 ± 0.01	-	0.01 ± 0.03	-	0.01 ± 0.03	-	0.04 ± 0.03	-
LBS	0.00 ± 0.00	-	0.00 ± 0.01	-	0.00 ± 0.00	-	0.00 ± 0.00	-	0.01 ± 0.00	-
MBRS	0.01 ± 0.04	-	0.01 ± 0.04	-	0.02 ± 0.04	-	0.01 ± 0.03	-	0.02 ± 0.02	-
RAW	-	0.20 ± 0.17	-	0.20 ± 0.17	-	0.20 ± 0.17	-	0.20 ± 0.17	-	0.20 ± 0.17
VINE	0.13 ± 0.00	-	0.12 ± 0.00	-	0.29 ± 0.00	-	-	-	-	-

TABLE X
RATCLIFF/OBERSHELP AND ACCURACY (MEAN ± STD) ACROSS METHODS AND MESSAGE LENGTHS. **ROTATION ATTACK**.

Method	Message length									
	2		4		8		15		32	
	Similarity score									
	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy
DWT-DCT	0.90 ± 0.30	-	0.87 ± 0.28	-	0.86 ± 0.28	-	0.88 ± 0.27	-	0.82 ± 0.28	-
LBS	0.02 ± 0.04	-	0.04 ± 0.06	-	0.05 ± 0.06	-	0.05 ± 0.05	-	0.06 ± 0.05	-
MBRS	1.00 ± 0.00	-	1.00 ± 0.00	-	1.00 ± 0.00	-	1.00 ± 0.00	-	1.00 ± 0.00	-
RAW	-	0.98 ± 0.10	-	0.98 ± 0.10	-	0.98 ± 0.10	-	0.98 ± 0.10	-	0.98 ± 0.10
VINE	0.98 ± 0.11	-	1.00 ± 0.00	-	1.00 ± 0.01	-	-	-	-	-

TABLE XI
RATCLIFF/OBERSHELP AND ACCURACY (MEAN ± STD) ACROSS METHODS AND MESSAGE LENGTHS. **SCALE ATTACK**.

Method	Message length									
	2		4		8		15		30	
	Similarity score									
	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy	Ratcliff	Accuracy
DWT-DCT	0.95 ± 0.22	-	0.92 ± 0.21	-	0.92 ± 0.21	-	0.93 ± 0.21	-	0.87 ± 0.25	-
LBS	1.00 ± 0.00	-	1.00 ± 0.00	-	1.00 ± 0.00	-	1.00 ± 0.00	-	1.00 ± 0.00	-
MBRS	1.00 ± 0.00	-	1.00 ± 0.00	-	1.00 ± 0.00	-	1.00 ± 0.00	-	1.00 ± 0.00	-
RAW	-	0.98 ± 0.10	-	0.98 ± 0.10	-	0.98 ± 0.10	-	0.98 ± 0.10	-	0.98 ± 0.10
VINE	0.97 ± 0.12	-	1.00 ± 0.00	-	1.00 ± 0.02	-	-	-	-	-

TABLE XII
RATCLIFF/OBERSHELP AND ACCURACY (MEAN ± STD) ACROSS METHODS AND MESSAGE LENGTHS. **NO ATTACK**.